

Week 1 Topic: Understand how energy costs affect affordability in NJ (residents + businesses)

1. What is the average monthly energy bill for a New Jersey household and for local businesses?

- **Residential Customers:**
 - \$127 per month
 - 19.88 cents per kWh
- **Commercial customers**
 - Grocery Stores
 - \$160 per month
 - \$4 sq ft per 50 kWh

2. How does this compare to other states?

New Jersey is *20% higher than national avg

- New York average electric bill- Rate: 22.19 cents, Bill: \$141.69 per month
NY average commercial electric bill- Rate: 19.23 cents, Bill: \$540 per month
- Connecticut average electric bill- Rate: 29.87 cents, Bill: \$190.38 per month
CT average commercial electric bill- Rate: 21.73 cents, Bill: \$580 per month
- Massachusetts average electric bill- Rate: 27.87 cents, Bill: \$153 per month
MA average commercial electric bill- Rate: 20.54 cents, Bill: \$1457 per month
- California average electric bill- Rate: 31.01 cents, Bill: \$163 per month
CA average commercial electric bill- Rate: 23.95 cents, Bill: \$1,199 per month

3. What are the biggest contributors to high costs (e.g., taxes, supply issues, infrastructure)?

- Aging infrastructure
- Taxes
- High population density
- Capacity Costs
- Supply delays (awaiting approval)
- Climate change

4. Explore how energy costs affect different groups—low-income families, small business owners, and large corporations.

- Low-income families:
 - Forced to choose between basic needs
 - Increasing budget strain on households
- *What is being done:
 - USF aims to reduce the energy burden to target levels for 90% of participating households.
 - Gov Murphy's relief plan- provide \$100 off electrical bills for all residents and \$250 for vulnerable families to combat high energy costs in the summer.
- Small business owners

- Increases operational expenses: electricity and fuel are major expenses including: manufacturing or transportation
- Price increases for customers: some businesses may be forced to increase their prices, thereby impacting consumer demand.

*actions being taken

- Governor Murphy's administration announced over \$430 million in direct relief to help lower energy bills for households and local businesses.
- The Board of Public Utilities (BPU) has required to develop proposals to lessen the impact of the recent rate hikes.
- Large corporations
 - Increased operational costs- businesses take a big setback due to the fact that energy and oil are the main sources for transportation.
 - Reduced Profitability- corporations will be forced to spend more from their own pockets, meaning banks will offer less loans or invest in the company.
 - Increased Prices- corporations will be forced to higher prices for consumers in order to offset high energy costs.
 - Relocation- businesses might be discouraged to expand within the state and might relocate to a state with lower energy costs.
 - Delayed Investment

Sources:

<https://www.saveonenergy.com/resources/electricity-bills-by-state/>
<https://www.eia.gov/state/print.php?sid=NJ>
<https://www.energybot.com/blog/average-business-electric-bill.html>
<https://poweroutage.us/electricity-rates>

Week 2 Topic: FACT SHEET as of June 12th, 2025

Topic: Wind Energy in NJ. What People Think: Public Support, Pushback, and Priorities

1. Break down New Jersey's offshore wind goals and where they actually stand:

Key Players	Role and Influence	Priorities
State Government	Governor Phil Murphy and NJ state government	11,000 megawatts (MW) of offshore wind capacity by 2040
Developers	Ørsted (Ocean Wind 1/2), Atlantic Shores	Ørsted cancelled 1.1 GW project in 2023 ; Atlantic Shores paused June 2025 amid permit & policy uncertainty
Federal Government	Trump Administration	Trump's 2025 executive order halted offshore wind permitting, stopping NJ activity
Climate Organizations	NJ Offshore Wind Alliance, Oceanic Network, Protect Our Coast NJ	Call for more jobs and carbon reduction. Against environmental

		impacts and aesthetics.
Local Officials/Citizens	Coastal Representatives (eg. Rep. Van Drew), Residents	Local backlash over tourism/whale concerns. Push for municipal control.

Project/developers	Power Capacity	Status	Delays/cancellations
Ocean wind 1 Ørsted	1,100 MW	Cancelled	- Rising costs (high inflation, interest rates) - supply chain delays - Project had been NJ's first awarded offshore wind farm - Faced vocal local opposition (concerns about tourism, fishing, environment)
Ocean wind 2 Ørsted	1,148 MW	Cancelled	- Rising costs (high inflation, interest rates) - supply chain delays - Project had been NJ's first awarded offshore wind farm - Faced vocal local opposition (concerns about tourism, fishing, environment)
Atlantic Shores Project 1 Shell ♦ EDF Renewables	1,510 MW	Terminated Contract	- Financial & regulatory hurdles: Inflation and supply-chain disruptions drove up costs. - Shell withdrew as a partner in Jan 2025, taking a \$996 million loss. - New federal actions in 2025 halted offshore wind

			permitting (President Trump's executive order) and led the EPA to suspend key permits, creating insurmountable delays. - NJ's regulators canceled the planned 2025 power contract solicitation due to "uncertainty driven by federal actions" and the loss of Shell.
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2. Support and Pushback for offshore wind energy among New Jersey politicians

Supporters:

- **Governor Phil Murphy**
 - **Executive Order 28** - established the goal of setting New Jersey on the path to 100% clean energy by 2050
 - **Executive Order #92** - raised New Jersey's offshore wind goal from 3,500 megawatts by 2030 to 7,500 megawatts by 2035
 - **Executive Order #307** - increased New Jersey's offshore wind goal by nearly 50 percent to 11,000 megawatts (MW) by 2040 - directs the New Jersey Board of Public Utilities to study the feasibility of increasing the target further
- **General Assembly Speaker Craig Coughlin - Statement on Offshore Wind Awards**
 - *"These awards put New Jersey back on track to be the national leader on offshore wind power"*
 - *"Climate change remains an existential threat to New Jersey's coastal communities and we should do our part to promote sustainable solutions with innovative projects like this."*

Opposition:

- **Congressman Chris Smith**
 - *"Two years ago the Congressman authored an amendment that passed the House of Representatives 244 to 189, requiring the GAO (Government Accountability Office) to comprehensively investigate the impact of offshore wind turbines off the coast of New Jersey and elsewhere."*
 - *"The offshore wind industrialization approval process has left unaddressed and unanswered numerous serious questions concerning the potentially harmful environmental impact on marine life and the ecosystems that currently allow all sea creatures great and small including whales to thrive."*

Congressman: Jeff Van Drew

- *"These offshore wind projects should have never been approved in the first place," Van Drew said in a statement, referring to President Joe*

Biden's effort to expand renewable energy as a "reckless green agenda that put politics over people."

- *We will fight tooth and nail to prevent this offshore wind catastrophe from wreaking havoc on the hardworking people who call our coastal towns home,"*

3. Reactions from New Jersey Communities

- Advocacy groups against offshore windmills have sprouted in coastal towns in New Jersey such as "Protect Our Coasts New Jersey"
- Issues communities have with offshore windmills: negatively affects tourism, fishing industry, harms wildlife, and is not the best environmentally sustainable option available (must be replaced, does not run at maximum power efficiency, etc.)

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- Positive reactions: would add jobs to New Jersey, would be predicted to supply up to 1 million homes per year

- Residents are having to pay higher electricity bills from around 17% to 20% in servicing areas.

Sources:

<https://www.njspotlightnews.org/2025/05/questioning-if-nj-offshore-wind-energy-development-dead-in-water/>

<https://www.heritage.org/energy/commentary/wind-turbines-not-green-not-reliable>

Week 3 Topic: Energy Price Trends in NJ (2017–2025): What's Driving Costs?

Components of Typical NJ Energy Bill

Usage: Amount of kilowatt-hours (kWh) used during the billing period. This varies due to the number of days in the billing period, family size, and general usage.

Meter Number: a unique number that identifies your electric meter at your home.

- Used by utility company to track your specific usage

Rate Schedule: the pricing plan assigned to your account that determines how much you are charged per kWh.

- Residential flat rate
- Time of use rate

Delivery Charge: Cover the cost of getting electricity to your home.

- Transmission charges: moving high voltage electricity from power plants to your area.
- Distribution charges: the local delivery to your house.
- Typically regulated by the NJ BPU

Supply Charges: cost of the electricity (price per kWh in NJ as of 2024 is \$0.13-\$0.16). This is set by the supplier and the biggest portion of the bill.

- Generation
- capacity

Average Daily Electric Use: 2023-2024 estimate would be 23-25 kWh/day. Factors that impact this include season, home size and insulation, and number of residents

Comparison Chart: comparing current use to last month, current use to same month last year, and average use to similar households.

June 2025 Energy Bill Increase

-Energy bills are estimated to increase 20.2% as of June 2025

-Energy bill increase comes from PJM's increased prices in their capacity market. PJM's prices increased because of the projected increase in demand for energy ([Ambrose 2025](#)).

-Demand has increased because of a shortage of new energy projects being built.

-The lack of energy projects being built has been attributed to NJP's poor planning. Other operators such as MISO have long-term plans that include implementing clean energy. MISO has plans which include scenarios with lower load growth and 40% decarbonization, and scenarios with higher load growth and 80% decarbonization ([Lang-Ree and McIntire 2025](#))

Estimated increase in future energy demand: ([Inside Lines 2025](#))

Year (Winter Season Dec – Feb)	Winter Peak (MW)	Change from 2024 Long-Term Load Forecast (MW/%)	Year	Summer Peak (MW)	Change From 2024 Long-Term Load Forecast (MW/%)
2024/25	136,127	-201 (-0.1%)	2025	154,144	651 (+0.4%)
2029/30	167,237	14,367 (+9.4%)	2030	183,883	16,010 (+9.5%)
2034/35	198,175	32,470 (+19.6%)	2035	209,923	30,301 (+16.9%)
2039/40	209,718	N/A	2040	220,224	N/A
2044/45	218,760	N/A	2045	228,544	N/A

New Jersey Energy Costs Compared to NJP States and Non-NJP States

<https://www.chooseenergy.com/electricity-rates-by-state/>

State	Residential Electricity Rates	Residential Electricity Rates	Annual percentage	National Rank
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	March 2025	March 2024	change	
Alabama	16.5 ¢/kWh	15.91 ¢/kWh	3.7	32
Alaska	25.79 ¢/kWh	24.71 ¢/kWh	4.4	44
Arizona	15.19 ¢/kWh	14.94 ¢/kWh	1.7	28
Arkansas	12.56 ¢/kWh	12.9 ¢/kWh	-2.6	10
California	32.41 ¢/kWh	32.53 ¢/kWh	-0.4	48

Recommendations:

Outline: After researching energy costs and issues with energy production, we have come to three sources of energy that could be improved upon in New Jersey to improve energy costs and energy output;

Liquified Natural Gas:

- LNG is natural gas stored as a liquid for transportation or emergency storage. New Jersey uses LNG from on-site storage tanks during peak energy demand and is supplemented into pipelines.
- Affordability: New Jersey households save almost 62% per year by using natural gas (\$420 per year instead of \$1,100 per year) ([U.S. Energy Information Administration](#)) ([American Gas Association](#)).
- In 2024, prices for residential natural gas averaged \$13.87 per thousand cubic feet, which was %7.1 less than other states ([Natural Gas Local](#)).

Nuclear Power:

- Increase Nuclear Energy Output → Construction of more nuclear reactors/employ use of Small modular reactors (SMRs)
- Pass Bill A5517 (Assemblyman Cody Miller, D-Gloucester), → Preamble: “this bill directs the Board of Public Utilities (board), in consultation with the Department of Environmental Protection and the New Jersey Economic Development Authority, to study, prepare, and submit, to the Governor and the Legislature, within 18 months of the effective date of this bill, a written report concerning the feasibility of developing advanced reactors

throughout the State and the feasibility of, need for, and design of a program in the State for developing small modular or advanced reactors”

- Remove the current moratorium on the construction of new nuclear power facilities → Education campaign on the safety and efficient of nuclear power
- “U.S. nuclear power plants are already among the safest and most secure industrial facilities in the world due to the industry’s commitment to comprehensive safety procedures, robust training programs and stringent federal regulation that keep nuclear plants and neighboring communities safe” - [US Department of Energy](#)

Small modular reactors (SMRs)

- SMRs are smaller scale nuclear plants that generate up to 300 Megawatts of energy compared to the regular level of 2,285 MW of the Salem nuclear power plants
- This technology offers the advantages of nuclear energy with lower costs & greater project flexibility

Concern: Implementation of SMR technology on a large scale would be a more of a long term project of the state. According to the Garden State Initiative, American SMRs won’t be operational until 2030.

Windmills: a renewable source with no fuel cost and low long-term maintenance

- NJ has strong wind potential
 - Aims for 7.5 GW of offshore wind capacity by 2035
- Types of windmills include offshore and onshore/community

Benefits:

- Lowers electricity costs over time
- Creates jobs in construction, maintenance, and manufacturing
- Reduces air pollution and dependence on fossil fuels

Challenges:

- High upfront costs
- Environmental concerns (impact on animals)

- Visual concerns from coastal communities

Recommendations:

- Secure long-term financing tools → Encourage federal-state collaboration to reinstate permitting paths once executive policies change.
 - Use state and federal subsidies to make wind power cheaper for households
- Strengthen local supply chains → Incentivize offshore wind component manufacturing in NJ ports (like Newark) to cut shipping costs and create jobs.

Water Conservation tips

- Shorten shower times → one to two minute showers can save up to 700 gallons per month
- Fixing leaky faucets → this can save 20 gallons per day for every leak stopped
- Using a broom instead of a hose → gutters, driveways, and sidewalks have to be cleaned, but a broom saves 150 gallons or more each time.